

Cody Pych

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COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Midsemester Exam #4

Only one sheet of paper 8 1/2 x 11 is allowed

No phones or i-pads or computers are allowed

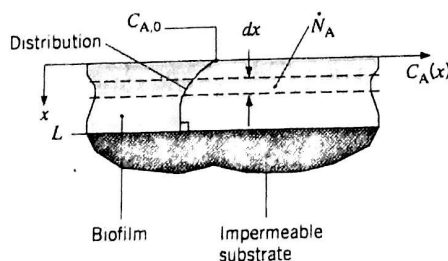
Duration: 75 minutes --- December 4, 2025

All Problems must be done

Problem 1: (35 points)

Biofilms, which are colonies of bacteria that can cling to living or inert surfaces, can cause a wide array of human infections. Infections caused by bacteria living within biofilms are often chronic because antibiotics that are applied to the surface of a biofilm have difficulty penetrating through the film thickness. Consider a biofilm that is associated with a skin infection. An antibiotic (species A) is applied to the top layer of a biofilm (species B) so that a fixed concentration of medication, $C_{A,0} = 4 \times 10^{-3} \text{ kmol/m}^3$, exists at the upper surface of the biofilm. The diffusion coefficient of the medication within the film, $D_{AB} = 2 \times 10^{-12} \text{ m}^2/\text{s}$. The antibiotic is consumed by biochemical reactions within the film, and the consumption rate depends on the local concentration of medication expressed as $\dot{N}_A = -k_1 C_A$ where $k_1 = 0.1 \text{ s}^{-1}$. To eradicate the bacteria, the antibiotic must be consumed at a rate of at least $0.2 \times 10^{-3} \text{ kmol/s} \cdot \text{m}^3$ ($\dot{N}_A \leq -0.2 \times 10^{-3} \text{ kmol/s} \cdot \text{m}^3$) since, at smaller absolute consumption rates, the bacteria will be able to grow back faster than it is destroyed. Determine the maximum thickness of a biofilm, L , that may be treated successfully by the antibiotic.

Schematic:



$$D_{AB} \frac{d^2 C_A}{dx^2} = -\dot{N}_A$$

$$\frac{d^2 C_A}{dx^2} = -\frac{\dot{N}_A}{D_{AB}}$$

$$\frac{d C_A}{dx} = -\frac{\dot{N}_A}{D_{AB}} x + A$$

$$C_A(x) = -\frac{\dot{N}_A}{2D_{AB}} x^2 + Ax + B$$

$$C_A(0) = 4 \times 10^{-3} = B$$

$$C_A(L) = 0.2 \times 10^{-3} = -\frac{\dot{N}_A}{2D_{AB}} L^2 + AL + 4 \times 10^{-3}$$

$$C_A(L) = 0.2 \times 10^{-3} = \frac{-A \pm \sqrt{A^2 - 4\left(-\frac{\dot{N}_A}{2D_{AB}}\right)(4 \times 10^{-3})}}{2\left(-\frac{\dot{N}_A}{2D_{AB}}\right)}$$

$$-\frac{\dot{N}_A}{2D_{AB}} = -\frac{-0.2 \times 10^{-3}}{2(2 \times 10^{12})} = 5 \times 10^{-17}$$

$$\frac{-A \pm \sqrt{A^2 - 8 \times 10^{-19}}}{1 \times 10^{-16}}$$

$$0.2 \times 10^{-20} = -A \pm \sqrt{A^2 - 8 \times 10^{-19}}$$

$$0.2 \times 10^{-20} + A = A^2 - 8 \times 10^{-19}$$

$$A = 8.02 \times 10^{-19}$$

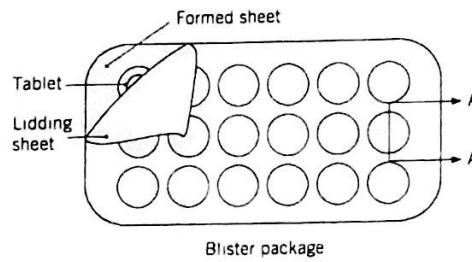
$$C_A(L) = .2 \times 10^{-3} = (5 \times 10^{-17}) L^2 + 8.02 \times 10^{-19} L + 4 \times 10^{-3}$$

$$= \frac{-8.02 \times 10^{-19} \pm \sqrt{(8.02 \times 10^{-19})^2 - 4(5 \times 10^{-17})(4 \times 10^{-3})}}{2(5 \times 10^{-17})}$$

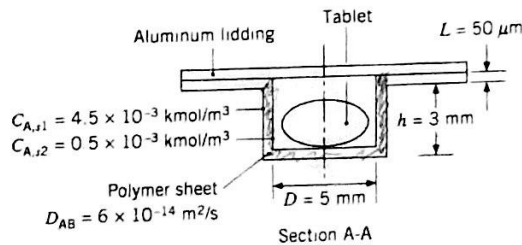
Problem 2: (35 points)

The efficacy of pharmaceutical products is reduced by prolonged exposure to high temperature, light, and humidity. For water vapor-sensitive consumer products that are in tablet or capsule form, and might be stored in humid environments such as bathroom medicine cabinets, *blister packaging* is used to limit the direct exposure of the medicine to humid conditions until immediately before its ingestion.

Consider tablets that are contained in a blister package composed of a flat *lidding sheet* and a second, *formed sheet* that includes troughs to hold each tablet. The formed sheet is $L = 50 \mu\text{m}$ thick and is fabricated of a polymer material. Each trough is of diameter $D = 5 \text{ mm}$ and depth $h = 3 \text{ mm}$. The lidding sheet is fabricated of aluminum foil. The binary diffusion coefficient for water vapor in the polymer is $D_{AB} = 6 \times 10^{-14} \text{ m}^2/\text{s}$ while the aluminum may be assumed to be impermeable to water vapor. For molar concentrations of water vapor in the polymer at the outer and inner surfaces of $C_{A,s1} = 4.5 \times 10^{-3} \text{ kmol/m}^3$ and $C_{A,s2} = 0.5 \times 10^{-3} \text{ kmol/m}^3$, respectively, determine the rate at which water vapor is transferred through the trough wall to the tablet.



Schematic:



$$D_{AB} \frac{d^2 C_A}{dx^2} = -R$$

$$\frac{d^2 C_A}{dx^2} = \frac{-R}{D_{AB}}$$

$$\frac{dC_A}{dx} = \frac{-R}{D_{AB}} x + A$$

$$C_A(x) = \frac{-R}{2D_{AB}} x^2 + Ax + B$$

$$C_{A,s1}(0) = 4.5 \times 10^{-3} = B$$

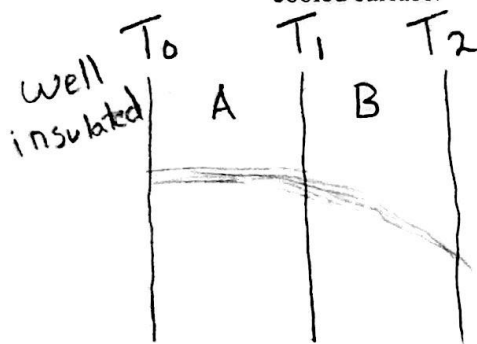
$$C_{A,s2}(5 \times 10^{-8}) = \frac{-R}{2D_{AB}} (5 \times 10^{-8})^2 + A(5 \times 10^{-8}) + 4.5 \times 10^{-3}$$

$$= 0.5 \times 10^{-3}$$

Problem 3: (35 points)

A plane wall is a composite of two materials, A and B. The wall of material A has uniform heat generation $\dot{q} = 1.5 \times 10^6 \text{ W/m}^3$, $k_A = 75 \text{ W/m} \cdot \text{K}$, and thickness $L_A = 50 \text{ mm}$. The wall material B has no generation with $k_B = 150 \text{ W/m} \cdot \text{K}$ and thickness $L_B = 20 \text{ mm}$. The inner surface of material A is well insulated, while the outer surface of material B is cooled by a water stream with $T_\infty = 30^\circ\text{C}$ and $h = 1000 \text{ W/m}^2 \cdot \text{K}$.

1. Sketch the temperature distribution that exists in the composite under steady-state conditions.
2. Determine the temperature T_0 of the insulated surface and the temperature T_2 of the cooled surface.



$$T_\infty = 30^\circ\text{C}$$

$$h = 1000$$

$$\frac{\text{W}}{\text{m} \cdot \text{K}} \quad \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$\dot{q} = 1.5 \times 10^6$$

$$\dot{q} = 0$$

$$L_A = 50 \text{ mm}$$

$$0.05 \text{ m}$$

$$L_B = 20 \text{ mm}$$

$$0.02 \text{ m}$$

$$K_A = 75$$

$$K_B = 150$$